

Sébastien Speierer

Staff Research Engineer – Meta Reality Labs Research

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Education

- 2015 – 2018 **EPFL**, Master's in Communication Systems – Lausanne, Switzerland
 - GPA: 5.29/6
- 2012 – 2015 **EPFL**, Bachelor in Communication Systems – Lausanne, Switzerland
 - GPA: 5.62/6
- 2014 – 2015 **Carnegie Mellon University**, Exchange Year in Computer Science – Pittsburgh, PA, USA
 - GPA: 3.52/4

Experience

- 2026 – present **Staff Research Engineer**, Meta - XR Graphics – Lausanne, Switzerland
 - Physics-based inverse rendering and neural rendering research
- 2023 – 2025 **Senior Research Engineer**, Meta - Reality Labs Research – Lausanne, Switzerland
 - Physics-based inverse rendering and neural rendering research
- 2019 – 2023 **Research Engineer**, Realistic Graphics Lab, EPFL – Lausanne, Switzerland
 - Development of the [Mitsuba 3](#) renderer and research in differentiable rendering, with Wenzel Jakob
- Jun – Oct 2019 **Visualization Software Engineer**, Blue Brain Project, EPFL – Geneva, Switzerland
 - Houdini pipeline development and differentiable rendering research
- 2018 – 2019 **Rendering Researcher**, Weta Digital – Wellington, New Zealand
 - Manuka production renderer team, research on volume rendering and reflectance filtering
- Jul – Sep 2017 **Rendering Researcher Intern**, Pixar Animation Studios – Emeryville, CA, USA
 - Research on combining many-lights sampling and path guiding strategies
- Feb – Dec 2017 **Research Assistant**, Realistic Graphics Lab, EPFL – Lausanne, Switzerland
 - Mitsuba 2 development
- Jul – Dec 2016 **Rendering Software Engineer Intern**, Pixar Animation Studios – Emeryville, CA, USA
 - Implementation of Manifold Next Event Estimation in RenderMan

Skills

Languages: C++, Python, CUDA, C, Java, Scala, Bash

Graphics: Mitsuba, Dr.Jit, PBRT, OptiX, Houdini, Blender, RenderMan, OpenGL, GLSL, Unity, Maya

Other: Git, PyTorch, Visual Studio, MATLAB, Mathematica, Photoshop, LaTeX, Windows, Linux

Awards and Film Credits

- Avatar: The Way of Water – Visual Effects, Weta Digital (2022)
- Avengers: Endgame – Visual Effects, Weta Digital (2019)
- Gemini Man – Visual Effects, Weta Digital (2019)
- 3rd best Bachelor's in Communication Systems at EPFL (2015)
- Carnegie Mellon University Dean's List (Spring 2015)

Languages and Interests

Languages: French (native), English (near-native)

Music: Classical piano degree; production, mixing and mastering with Cubase and Ableton Live

Publications and Projects

- ArXiv 2026 [Generalized Non-Exponential Gaussian Splatting](#)
Sébastien Speierer, Adrián Jarabo
- SIGGRAPH 2026 [Beyond Positional Encoding: A 5D Spatio-Directional Hash Encoding](#)
Philippe Weier, Lukas Bode, Philipp Slusallek, Adrián Jarabo, **Sébastien Speierer**
- SIGGRAPH 2026 [Don't Splat your Gaussians: Volumetric Ray-Traced Primitives for Modeling and Rendering Scattering and Emissive Media](#)
Jorge Condor, **Sébastien Speierer**, Lukas Bode, Aljaž Božič, Simon Green, Piotr Didyk, Adrián Jarabo
- SIGGRAPH 2025 [Image-space Adaptive Sampling for Fast Inverse Rendering](#)
Kai Yan, Cheng Zhang, **Sébastien Speierer**, Guangyan Cai, Yufeng Zhu, Zhao Dong, Shuang Zhao
- EGSR 2025 [Controllable Biophysical Human Faces](#)
Minghao Liu, Stephane Grabli, **Sébastien Speierer**, Nikolaos Sarafianos, Lukas Bode, Matt Chiang, Christophe Hery, James Davis, Carlos Aliaga
- 2019 – present [Mitsuba 3](#)
Rendering system for forward and inverse light transport simulation
- SIGGRAPH 2022 [Dr.Jit: A Just-In-Time Compiler for Differentiable Rendering](#)
Wenzel Jakob, **Sébastien Speierer**, Nicolas Roussel, Delio Vicini
- SIGGRAPH 2022 [Differentiable Signed Distance Function Rendering](#)
Delio Vicini, **Sébastien Speierer**, Wenzel Jakob
- SIGGRAPH 2021 [Monte Carlo Estimators for Differential Light Transport](#)
Tizian Zeltner, **Sébastien Speierer**, Iliyan Georgiev, Wenzel Jakob
- SIGGRAPH 2021 [Path Replay Backpropagation: Differentiating Light Paths using Constant Memory and Linear Time](#)
Delio Vicini, **Sébastien Speierer**, Wenzel Jakob
- CVPR 2021 [Wide-Depth-Range 6D Object Pose Estimation in Space](#)
Yinlin Hu, **Sébastien Speierer**, Wenzel Jakob, Pascal Fua, Mathieu Salzmann
- Patent 2021 [Method for improved handling of texture data for texturing and other image processing tasks](#)
Kimball D. Thurston III, Luca Fascione, **Sébastien Speierer**
- SIGGRAPH 2020 [Radiative Backpropagation: An Adjoint Method for Lightning-Fast Differentiable Rendering](#)
Merlin Nimier-David, **Sébastien Speierer**, Benoît Ruiz, Wenzel Jakob
- Thesis 2018 [Spatially-varying Specular Microstructures and Reflectance Filtering in a Production Renderer](#)
Sébastien Speierer, Andrea Weidlich, Wenzel Jakob
- Pixar 2018 [Caustic Connection Strategies for Bidirectional Path Tracing](#)
Sébastien Speierer, Christophe Hery, Ryusuke Villemin, Wenzel Jakob
- EPFL 2016 [Metropolis Virtual Point Light Rendering](#)
Sébastien Speierer, Wenzel Jakob